Transformation Geometry — problem 12.2.8

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Consider n circles C_i ($C_{n+1} = C_1$) with C_i touching C_{i+1} externally at T_i for $i = 1$ to n. Start at any point A_1 on C_1, and, for $i = 1$ to n, draw straight lines A_iT_i intersecting C_{i+1} a second time in A_{i+1}. What is the relative position of A_1 and A_{n+1} on C_1 ? Generalize.